

The Fertility, Marriage, and Gender Equality Quandary

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Motivation

- Three trends underpin the grand gender convergence in the last century:
 1. [Falling fertility](#) (Guinnane 2011)
 2. [Declining dual parenthood](#) (Stevenson and Wolfers 2007)
 3. [Converging gender \(income\) gaps](#) (Goldin 2014)
- Many policies target [individual](#) outcomes (pro-natalism, pro-marriage, pro-equality)
- Existing research treats outcomes in isolation or pairwise, lacking a unified framework

This Paper

1. Document a **three-way trade-off**: simultaneously achieving high fertility, low single parenthood, and gender income equality is statistically rare
⇒ deep dependency structure, potential policy boundaries
2. Develop a **unified equilibrium model** of marriage, fertility, and female labor supply with two-dimensional marriage market pricing
3. Show level policies improve two at the expense of the third; **redistributing childcare** can loosen the trade-off under transparent conditions
4. Calibrate to Mexico (1990–2015): gender-neutral and gender-biased TFP
⇒ decomposition, welfare analysis, model-based predictions

Roadmap

- Empirical findings
- Model and policy analysis
- Quantitative application: Mexico
- Conclusion

Four Datasets to Document the Three-Way Trade-Off

	OECD	World	U.S. States	U.S. CZ
Coverage	37 countries	95 countries	50 + D.C.	741 CZs
Period	1970–2014	1990–2015	2023	2000
Obs.	721	1,130	51	741
Fertility	TFR	TFR	Births / 1000	TFR
Dual parenthood	Out-of-wedlock	Both parents	Single-mother	Single-mother
Gender gap	Earnings	Income share	Earnings	Earnings
Advantage	Long horizon	Wide coverage	Uniform inst.	Granular

- Construct dummies at sample medians: high F, dual M, equal G
- **Dartboard benchmark**: under independence, trinity share is $0.5^3 = 12.5\%$

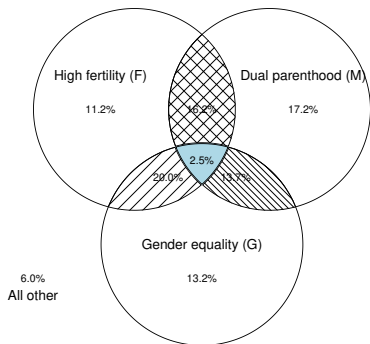
The Trinity is Rare

OECD

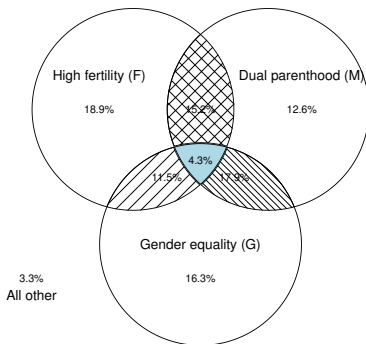
World

U.S.

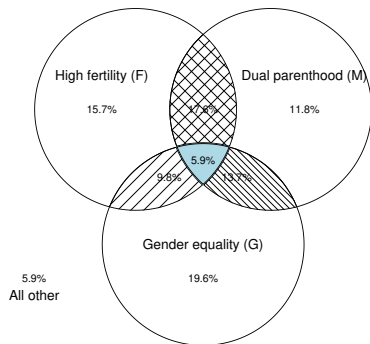
synergy



OECD



World



U.S. States

Model: Preferences

- Building on Becker (1973) and Choo and Siow (2006); gender $g \in \{\sigma, \varphi\}$
- Preferences over consumption c and number of children n :

$$u(c, n) = \left((1 - \beta) c^{\frac{\rho-1}{\rho}} + \beta n^{\frac{\rho-1}{\rho}} \right)^{\frac{\rho}{\rho-1}}, \quad \rho > 1$$

so that $u(c, 0)$ is well-defined

- Homogeneous wages within gender; gender wage gap $\Gamma^w = w^\sigma / w^\varphi$

Model: Men's Problem

- If single, men consume labor income but have no children:

$$V^{\sigma^{\rightarrow},s} = u(w^{\sigma^{\rightarrow}}, 0)$$

- Once married, husbands transfer a share α of income to wives:

$$V^{\sigma^{\rightarrow},m} = u((1 - \alpha)w^{\sigma^{\rightarrow}}, n^m) \cdot \lambda$$

- $\alpha \in [-w^{\sigma^{\rightarrow}}/w^{\sigma^{\rightarrow}}, 1]$ is endogenous in the marriage market equilibrium
- n^m is the household fertility level, also endogenous

Model: Women's Problem

- Single women solve

$$V^{\text{♀},s} = \max_{c,l,n} u(c, n) \quad \text{s.t.} \quad c = w^{\text{♀}}l, \quad l = 1 - \chi n$$

- Married women solve

$$V^{\text{♀},m} = \max_{c,l,n} u(c, n) \cdot \lambda \quad \text{s.t.} \quad c = \alpha w^{\text{♂}} + w^{\text{♀}}l, \quad l = 1 - \chi n$$

- Trade-off between consumption and fertility through female labor supply
- In the baseline, wives bear all childcare; results hold so long as $\chi^{\text{♂}} < \chi^{\text{♀}}$

Marriage Market Equilibrium

- Marriage rates from Fréchet shocks:

$$\mathcal{M}^{\sigma} = \frac{1}{1 + (V^{\sigma,s}/V^{\sigma,m})^{\theta}}, \quad \mathcal{M}^{\varphi} = \frac{1}{1 + (V^{\varphi,s}/V^{\varphi,m})^{\theta}}$$

- Equilibrium: $\mathcal{M}^{\sigma} = \mathcal{M}^{\varphi} = \mathcal{M}$
- **Two-dimensional price** (α, n^m) : marriage bundles income- and fertility-sharing
- Lemma 1: Given $\{w^{\sigma}, w^{\varphi}\}$, the equilibrium (α, n^m) is unique
- Lemma 2: $V^{\varphi,m}/V^{\varphi,s} = \lambda(1 + \alpha\Gamma^w)$

Three-Way Trade-Off and Level Policies

- Equilibrium tension:

$$\mathcal{M} = \frac{(\lambda(1 + \alpha\Gamma^w))^{\theta}}{1 + (\lambda(1 + \alpha\Gamma^w))^{\theta}}, \quad \Gamma^y = \Gamma^w / l^{\varnothing}, \quad l^{\varnothing} = 1 - \chi n$$

- Insight:** Higher n reduces $l^{\varnothing}$, requiring low Γ^w for gender income equality—but low Γ^w reduces $\mathcal{M}$.

Policy	Fertility (n)	Marriage ($\mathcal{M}$)	Gender Gap (Γ^y)
Pro-marriage ($\uparrow \lambda$)	+	+	+ (widens)
Gender equality ($\uparrow w^{\varnothing}$)	–	–	– (narrows)
Reduce child costs ($\downarrow \chi$)	+	+	+ (widens)

- Extension: $\chi^{\text{♀}}$ for wives, $\chi^{\text{♂}}$ for husbands
- Counterfactual: redistribute δ from wives to husbands
- **Proposition:** Redistribution improves all three outcomes if:
 1. Wife's earnings exceed spousal transfers: $w^{\text{♀}} > \alpha w^{\text{♂}}$
 2. Husband's gain from fertility exceeds consumption loss:

$$\frac{\beta}{1-\beta} \left(\frac{c^{\text{♂},m}}{n^m} \right)^{1/\rho} > (1-\alpha)w^{\text{♂}} \left(\frac{n^m}{\partial n^m / \partial \delta} + \chi^{\text{♂}} \right)$$

More likely if β is large, baseline $\chi^{\text{♂}}$ is small, fertility response is strong

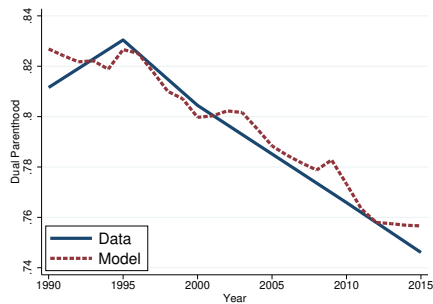
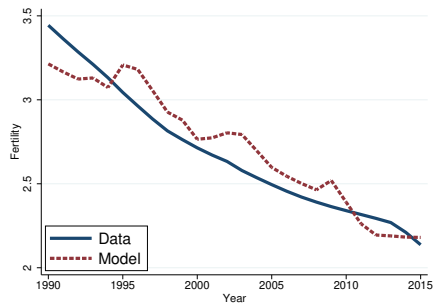
- When both hold, redistribution is **Pareto-improving**

Mexico 1990–2015: Data and Calibration

trends

tech

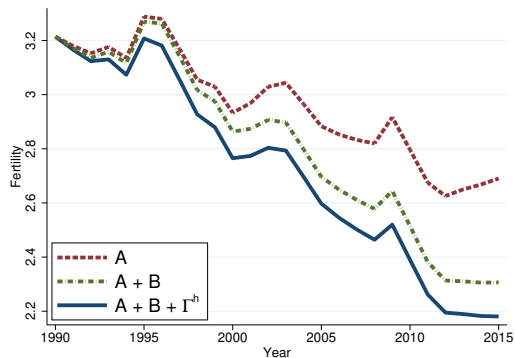
- Decompose wages: $A_t = w_t^{\sigma^{\nearrow}}$ (gender-neutral), Γ_t^h (human capital gap), $B_t = (w_t^{\sigma^{\nearrow}} / w_t^{\sigma^{\searrow}}) / \Gamma_t^h$ (gender-biased)
- Fix $\chi = 0.075$ (de La Croix & Doepke 2003), $\chi^{\sigma^{\nearrow}} \approx 0$; calibrate $\beta = 0.104$, $\rho = 1.50$, $\lambda = 1.03$, $\theta = 3.40$



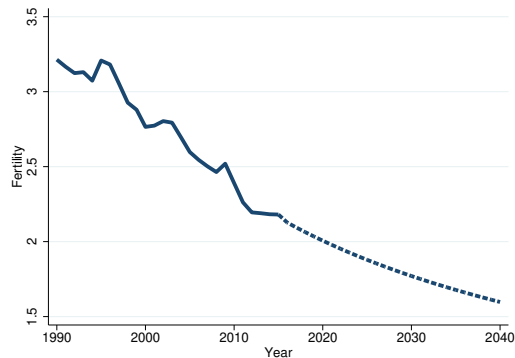
Mexico: Drivers and Out-of-Sample Test

full decomp

α , welfare



Fertility decomposition



2023 prediction: 1.93 vs. actual 1.91

- Gender-neutral TFP explains ~50% of falling fertility and marriage
- Gender-biased TFP and education reversal drive income convergence

Conclusion

- A three-way trade-off with synergistic dependency
- Unified framework: two-dimensional marriage-market pricing
- Level policies cannot escape the trade-off; redistribution of childcare can—under checkable conditions
- Mexico: gender-neutral and gender-biased TFP both matter; strong out-of-sample fit
- Dynamic propagation through male human capital

dynamic extension

Synergy, not Redundancy

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- Mutual information decomposition: pairwise correlations *understate* joint structure
- **Synergy**: conditioning on any one variable *strengthens* the dependence between the other two

Sample	Total Corr.	Interaction Information	Interpretation
OECD	0.165	−0.035***	Synergy
World	0.169	−0.026***	Synergy
U.S. States	0.112	−0.007	Synergy

Example (OECD): $I(F; M) = 0.043$ bits, but $I(F; M | G) = 0.078$ bits—**nearly double**

⇒ Calls for a unified framework rather than three separate stories

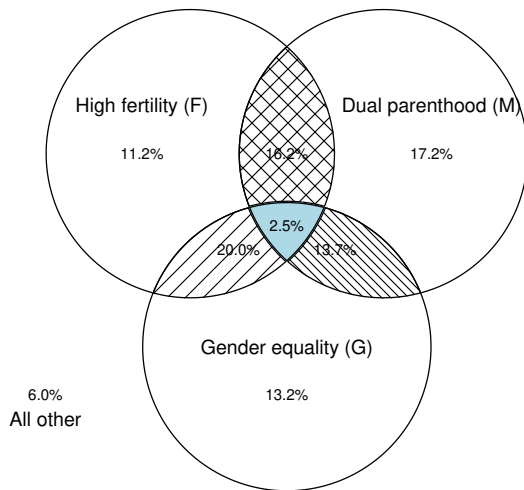
OECD Sample

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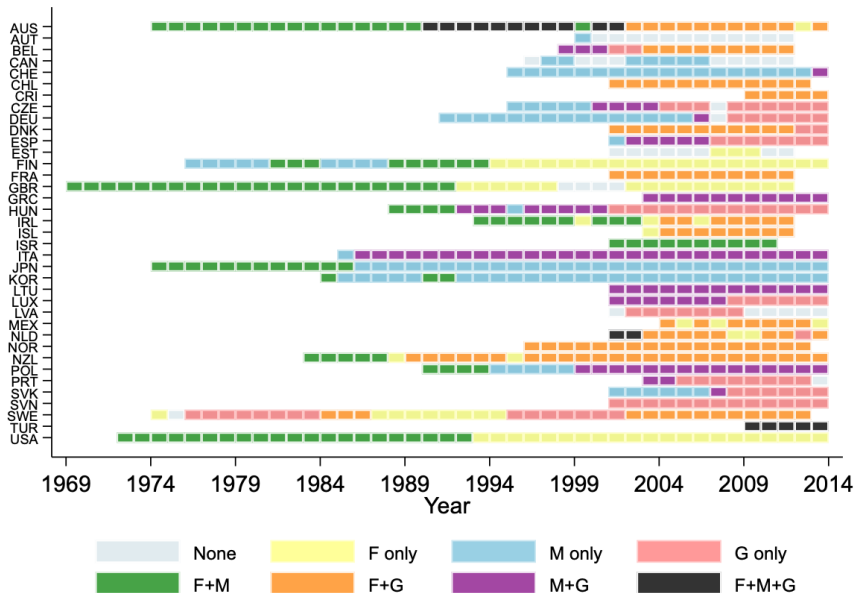
37 OECD countries, 1970–2014
721 observations

- Total fertility rate (1.69)
- Out-of-wedlock birth share (31.4%)
- Gender median earnings gap (17.2%)

Advantage: longest horizon



OECD Sample – Transition Paths

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World Sample

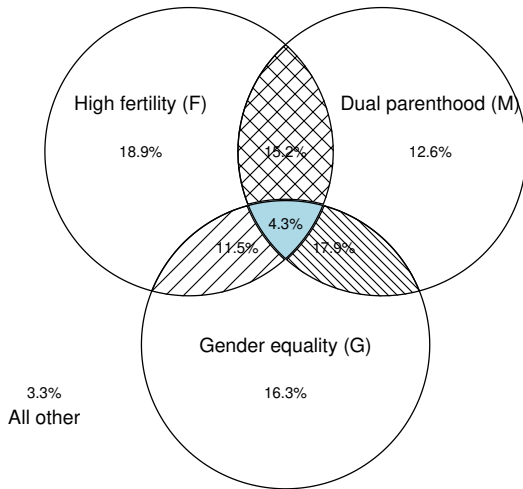
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95 countries, 1990–2015

1,130 observations

- Total fertility rate (2.81)
- Children with both parents (75.9%)
- Female labor income share (32.3%)

Advantage: better measurements, wider coverage



U.S. Samples

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States: 50 + D.C., 2023

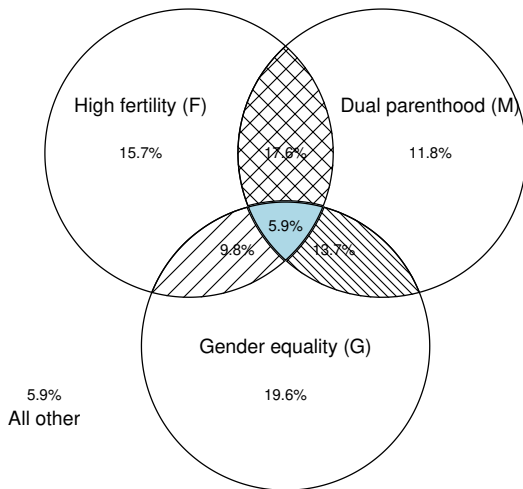
- Births per 1000 women (56.9)
- Single-mother share (17%)
- Gender earnings gap (25.6%)

Commuting Zones:

741 CZs, 2000

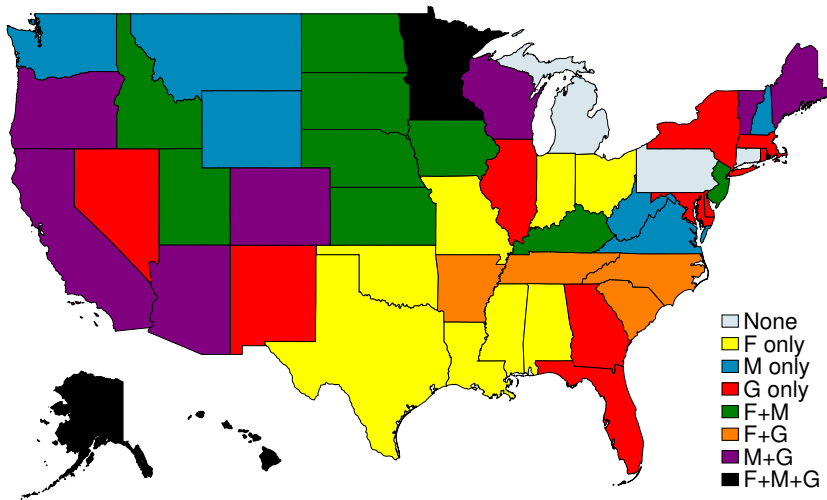
Trinity share: 7.6%

Advantage: uniform institutional setting



U.S. States – Geographic Distribution

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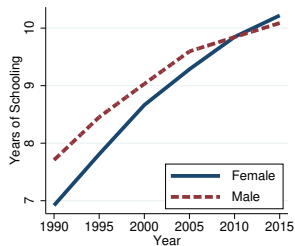
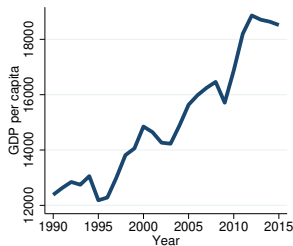
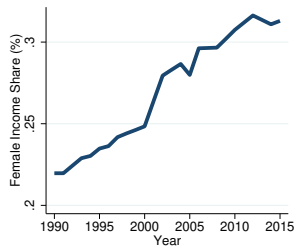
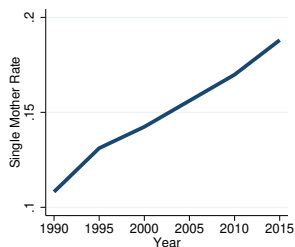
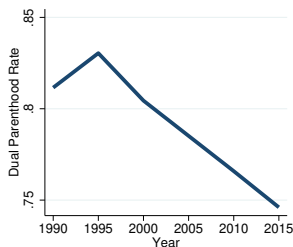
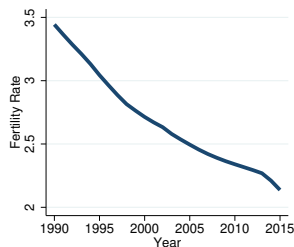
Three states achieve the trinity: [Alaska](#), [Hawaii](#), [Minnesota](#)

Organizing the Paternity-Leave Evidence

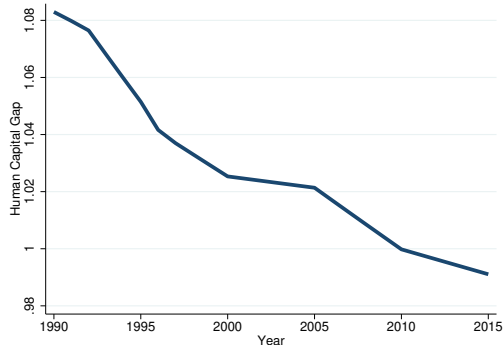
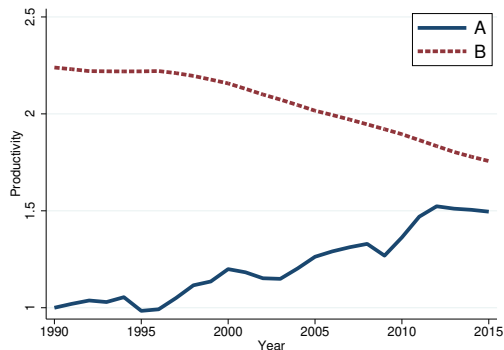
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- **Gender-gap channel (robust)**: reserved quotas narrow within-household division of childcare in Spain, Germany, Norway, Quebec
- **Fertility channel (heterogeneous)**:
 - Null/negative in Spain, Norway (high-wage, low-fertility; modest β , higher baseline χ^{σ^*})
 - Positive in Quebec, low-baseline χ^{σ^*} subgroups in Germany
- Pattern tracks condition (2) of the proposition
- **Falsifiable prediction**: in middle-income settings (high β , low χ^{σ^*} , moderate w^{φ}), reforms should raise fertility *and* narrow gender gaps

Mexico's Transition (1990–2015)

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Wage Decomposition

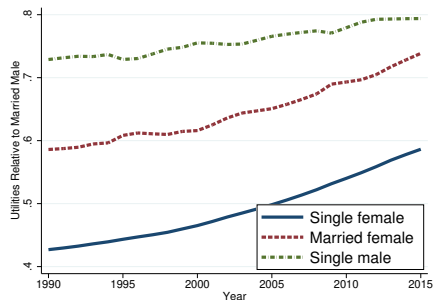
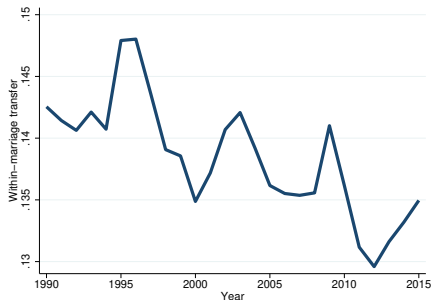
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Left: gender-neutral TFP A_t and gender-biased component B_t . Right: human capital gap Γ_t^h

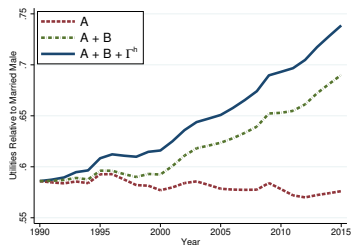
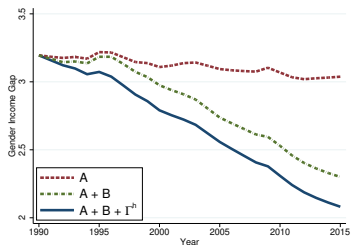
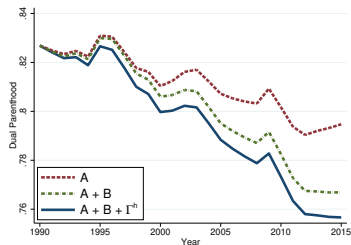
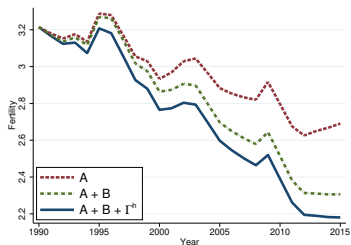
Equilibrium Outcomes: α and Welfare

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- α_t falls only modestly despite sharp decline in n^m : narrowing Γ^w requires larger share of smaller relative income
- Husband-wife welfare gap shrinks from 42% to 26% (driven by rising w^{f} , not larger transfers)
- $\alpha \approx 0.10$ satisfies condition (1); condition (2) also holds throughout



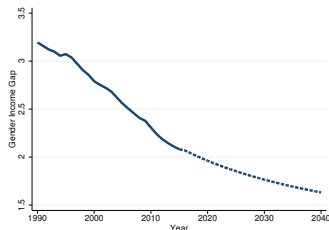
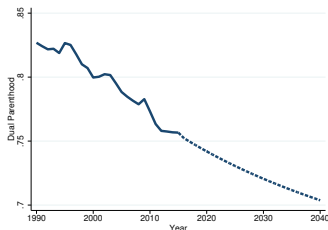
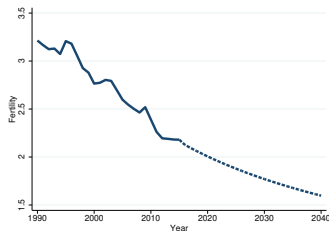
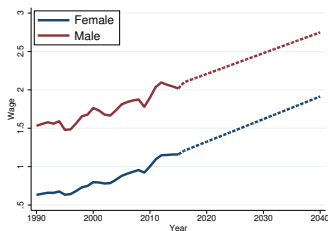
Full Decomposition

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Out-of-Sample Predictions

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- Extrapolate wages linearly from 1990–2015 trend
- 2023: model predicts fertility of 1.93; actual is 1.91



Dynamic Extension: Endogenous Human Capital Gap [back](#)

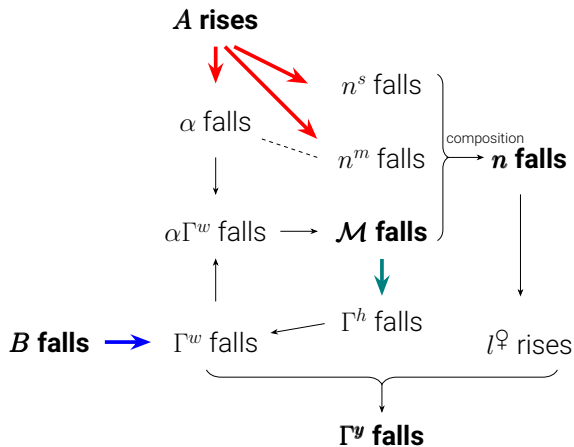
- Single parenthood affects boys and girls differently (Bertrand & Pan 2013, Autor et al. 2019, 2023, Wasserman 2020, Reeves 2022)

- Law of motion:

$$\Gamma_{t+1}^h = \mathcal{H}(\mathcal{M}_t), \quad \mathcal{H}'(\mathcal{M}_t) > 0$$

- Self-reinforcing feedback: declining marriage $\rightarrow$ lower male human capital $\rightarrow$ further marriage decline [diagram](#)

Dynamic Mechanism

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Red: A_t rises Blue: B_t falls Teal: Dynamic feedback